[1]:
```python
# Program to find squares using map()

numbers = [1, 2, 3, 4, 5]

squares = list(map(lambda x: x * x, numbers))

print("Original List:", numbers)
print("Squares List:", squares)
```

```
Original List: [1, 2, 3, 4, 5]
Squares List: [1, 4, 9, 16, 25]
```

[2]:
```python
# Program to filter even numbers

numbers = [10, 15, 20, 25, 30, 35, 40]

even_numbers = list(filter(lambda x: x % 2 == 0, numbers))

print("Original List:", numbers)
print("Even Numbers:", even_numbers)
```

```
Original List: [10, 15, 20, 25, 30, 35, 40]
Even Numbers: [10, 20, 30, 40]
```

[3]:
```python
# Program to find sum using reduce()

from functools import reduce

numbers = [1, 2, 3, 4, 5]

sum_result = reduce(lambda x, y: x + y, numbers)

print("List Elements:", numbers)
print("Sum of Elements:", sum_result)
```

```
List Elements: [1, 2, 3, 4, 5]
Sum of Elements: 15
```

[4]:

```python
# Program demonstrating higher-order function

def multiplier(n):
    return lambda x: x * n

double = multiplier(2)
triple = multiplier(3)

print("Double of 5:", double(5))
print("Triple of 5:", triple(5))
```

```
Double of 5: 10
Triple of 5: 15
```

[5]:

```python
# Program to sort tuples using lambda

students = [
    ("Anu", 85),
    ("Rahul", 72),
    ("Priya", 90),
    ("Kiran", 78)
]

sorted_students = sorted(students, key=lambda x: x[1])

print("Students Sorted by Marks:")
for student in sorted_students:
    print(student)
```

```
Students Sorted by Marks:
('Rahul', 72)
('Kiran', 78)
('Anu', 85)
('Priya', 90)
```

[6]:
```python
# Program to convert strings to uppercase using map()

names = ["anu", "rahul", "priya", "kiran"]

uppercase_names = list(map(lambda x: x.upper(), names))

print("Original Names:", names)
print("Uppercase Names:", uppercase_names)
```

```
Original Names: ['anu', 'rahul', 'priya', 'kiran']
Uppercase Names: ['ANU', 'RAHUL', 'PRIYA', 'KIRAN']
```

[7]:
```python
# Program to filter positive numbers

numbers = [-10, 20, -30, 40, 50, -60]

positive_numbers = list(filter(lambda x: x > 0, numbers))

print("Original List:", numbers)
print("Positive Numbers:", positive_numbers)
```

```
Original List: [-10, 20, -30, 40, 50, -60]
Positive Numbers: [20, 40, 50]
```

[8]:
```python
# Program to find maximum number using reduce()

from functools import reduce

numbers = [12, 45, 67, 23, 89, 34]

maximum = reduce(lambda x, y: x if x > y else y, numbers)

print("List Elements:", numbers)
print("Maximum Element:", maximum)
```

```
List Elements: [12, 45, 67, 23, 89, 34]
Maximum Element: 89
```

```
Maximum Element: 89
```

[9]:
```python
# Program using lambda function for addition

addition = lambda a, b: a + b

num1 = 10
num2 = 20

result = addition(num1, num2)

print("First Number:", num1)
print("Second Number:", num2)
print("Sum:", result)
```

```
First Number: 10
Second Number: 20
Sum: 30
```

[10]:
```python
# Program to calculate length of strings using map()

words = ["Python", "Java", "C", "JavaScript"]

lengths = list(map(lambda x: len(x), words))

print("Words:", words)
print("Lengths:", lengths)
```

```
Words: ['Python', 'Java', 'C', 'JavaScript']
Lengths: [6, 4, 1, 10]
```